

Task 8 – Random Forest vs Decision Tree

Objective:

Implement an ensemble model using Random Forest and compare it with Decision Tree.

Tools Used:

Python, Pandas, Scikit-learn, Matplotlib

Step 1 – Data Loading: Breast Cancer dataset was loaded and split into training and testing sets.

Step 2 – Decision Tree: A Decision Tree classifier was trained and evaluated for prediction accuracy.

Step 3 – Random Forest: Random Forest with 100 trees was trained. This model combines multiple decision trees to reduce overfitting and improve accuracy.

Step 4 – Accuracy Comparison: Random Forest achieved higher accuracy than Decision Tree because ensemble models reduce variance.

Step 5 – Feature Importance: Random Forest provides feature importance which helps identify the most influential features.

Interview Questions:

1. What is Bagging?

Bagging combines multiple models trained on random subsets of data to reduce variance.

2. How does Random Forest select features?

Random Forest selects a random subset of features at each split when building trees.

Conclusion:

Random Forest performs better than a single Decision Tree and is more robust.